

App Dev Project Report

1. Student Details

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About Me: I have completed my B.Tech in Information Technology and will be pursuing an M.Tech in Artificial Intelligence Systems. I am passionate about full-stack web development, artificial intelligence, and building data-driven applications that solve real-world problems.

2. Project Details

Project Title: Placement Portal

Problem Statement:

Create and design an user-friendly Placement Portal Application for campus placement in order to make the recruitment process more efficient, which will enable the administrators, companies and students to interact on a single platform. The system will help to streamline company approvals, placement drives, student applications and recruitment processes.

Approach:

The backend is built with Flask, the frontend is made using Vue.js, the data is stored in SQLite, asynchronous and scheduled tasks are supported with Celery, and Redis is used as cache. The system was designed to be secure, scalable, and user-friendly, with the following features being implemented: The following features were implemented in the system to ensure its security, scalability, and user-friendliness

3. AI/LLM Declaration

I used ChatGPT (OpenAI GPT-5) to assist with understanding concepts, debugging issues, designing the project architecture, generating documentation, refining the user interface, and resolving integration problems involving Flask, Vue.js, Redis, Celery, and JWT authentication. The extent of AI/LLM usage was approximately **20–25%**, primarily for guidance, code suggestions, and documentation.

All implementation, integration, testing, debugging, and final validation of the project were carried out manually by me.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework for developing RESTful APIs.
Vue.js 3	Frontend framework for building the user interface.
SQLite	Lightweight relational database for storing application data.
SQLAlchemy	Object Relational Mapper (ORM) for database operations.
Bootstrap 5	Frontend styling and responsive design.
JWT (Flask-JWT-Extended)	User authentication and role-based authorization.
Redis	In-memory caching to improve application performance.
Celery	Executes asynchronous and scheduled background jobs.
Flask-Mail & MailHog	Email service and local email testing for reminders and reports.

5. Database Schema / ER Diagram

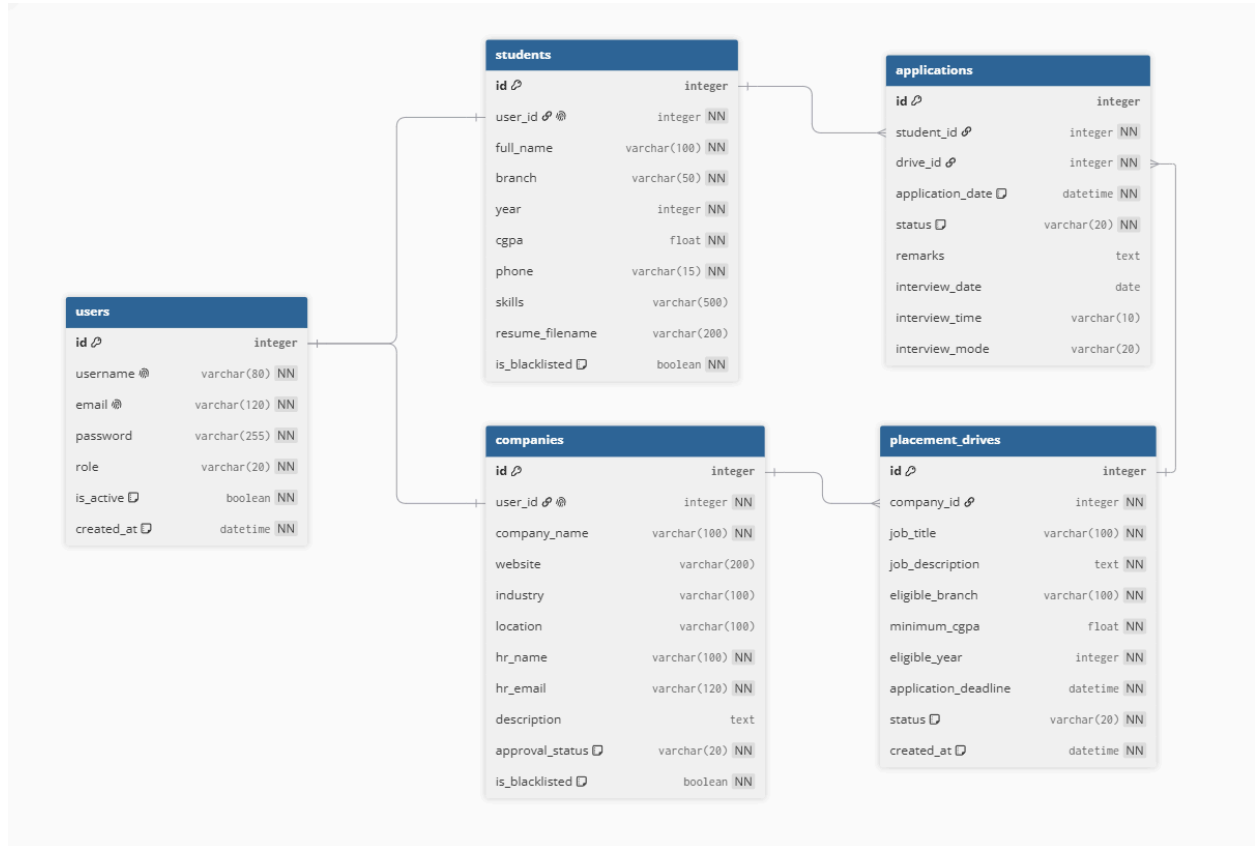
Tables

- **Users** – Stores login credentials and role information for Admin, Students, and Companies.
- **Students** – Stores student profile, academic details, skills, and resume information.
- **Companies** – Stores company profile, HR details, and approval status.
- **Placement_Drives** – Stores placement drive details and eligibility criteria for recruitment.
- **Applications** – Stores student applications, recruitment status, and interview details.

Relationships

- **One-to-One** → **Users** → **Students**
- **One-to-One** → **Users** → **Companies**
- **One-to-Many** → **Companies** → **Placement_Drives**
- **One-to-Many** → **Students** → **Applications**

- **One-to-Many** → **Placement_Drives** → **Applications**



6. API Resource Endpoints.

Endpoint	Method	Description
/auth/login	POST	Authenticates users and returns a JWT access token.
/admin/dashboard	GET	Retrieves dashboard statistics for the admin.
/student/drives	GET	Returns approved placement drives for students.
/student/apply/{drive_id}	POST	Allows a student to apply for a placement drive.
/company/drives	POST	Creates a new placement drive for admin approval.

YAML API Definition File: Included separately in the submission ZIP as [api.yaml](#).

7. Architecture and Features (optional)

Architecture Overview:

- app.py – Main file to run the Flask application.
- models / Database models using SQLAlchemy.
- routes/ – REST API endpoints (Flask Blueprints).
- services/ – Business logic, caching, email & background jobs.
- frontend/ – Vue.js front-end interface using the Bootstrap styling.

Implemented Features

- User Authentication and Role Based Access Control based on JWT.
- Placement Drive Management by Company Registration.
- Application forms of the students, uploading of their resumes, and their placement records.
- Application monitoring, search and approvals on admin dashboard.
- Redis Caching, Celery Background Jobs, MailHog email testing.

Additional Features

- Reminder emails sent daily and activity report sent monthly.
- Asynchronous student and company CSV export.
- Scheduling of interviews and tracking of the status of applications.

8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1MfQ1SnPMTU4hgqPyAUAsDYGJRZbqws42/view?usp=sharing>